



SARDAR PATEL INSTITUTE OF TECHNOLOGY

MID SEMESTER EXAMINATION

Sr.No. 8653

AV: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C Code: C5413
Exam Type: Written	Block:605	Seat No: 2022300092
	Course Section:1	



Name of the candidate. Sneha Sanjay Sambare.

Candidate's UID number:

2	0	2	2	3	0	0	0	9	2
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Examination class room number:

5	0	8
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Invigilator's signature with date:

npw 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTech

(For example FY MCA/FY BTECH)

SEMESTER : V\I

PROGRAM:.....Computer.....Engineering

I / II / III / IV / V / VI / ~~VII~~ / VIII

DAY AND DATE:.....Wednesday, 08/10/2025.....

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



START WRITING HERE
(Begin answer for each question on a new page)

Space
For
Marks

Q.1

a

$$f(w) = \frac{1}{2} \|xw - y\|^2$$

$$= \frac{1}{2} (xw^2 - y^2)$$

$$= \frac{1}{2} xw^2 - \frac{1}{2} y^2$$

Partial derivatives of $f(x)$ w.r.t. x & y

$$\frac{\partial f}{\partial xw} = xw - \frac{1}{2} y^2 = xw - \frac{1}{2} y^2$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} xw^2 - y = \frac{1}{2} xw^2 - y$$

$$\frac{\partial f}{\partial w} = xw, \quad \frac{\partial f}{\partial y} = y$$

Double partial derivatives

$$\frac{\partial^2 f}{\partial w^2} = xw = x - \frac{1}{2} y^2$$

$$\frac{\partial^2 f}{\partial y^2} = y = 1 - \frac{1}{2} xw^2$$

$$\frac{\partial^2 f}{\partial x \partial y} = xw - \frac{1}{2} y^2 = xw - y$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{1}{2} xw^2 - y = xw - y$$

Hessian matrix =

$$\begin{bmatrix} x - \frac{1}{2} y^2 & xw - y & xw - y & \frac{1}{2} xw^2 \\ x & xw - y & xw - y & 1 \end{bmatrix}$$

$$V_{k+1} = V_k - \eta \nabla f(w; y) \quad \text{where } V = (w, y)$$

$$\left[\left(x - \frac{1}{2} y^2 \right) - \lambda \right] (xw - y) (xw - y) \left[\left(\frac{1}{2} xw^2 \right) - \lambda \right]$$

Space
For
MarksQuestion
No.

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(Begin answer for each question on a new page)

$$= \begin{bmatrix} x & x\omega - y & x\omega - y & 1 \end{bmatrix}$$

eigen values :-

$$\det(H - \lambda I) = 0$$

$$\begin{bmatrix} x - \lambda & x\omega - y & x\omega - y & 1 - \lambda \end{bmatrix}$$

$$(x - \lambda) - (x\omega - y)(x\omega - y) - (1 - \lambda) = 0$$

$$(x - \lambda)(1 - \lambda) - [x\omega^2 - x\omega y + y^2 - x\omega y] = 0$$

$$[x - x\lambda - \lambda + \lambda^2] - [x\omega^2 - x\omega y + y^2 + x\omega y] = 0$$

$$[x - x\lambda - \lambda + \lambda^2] - [x\omega^2 + y^2] = 0$$

$$a = 1, b = -1, c = 1$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{1 - 4}}{2}$$

$$= \frac{-1 + \sqrt{5}}{2}$$

$$= \frac{-1 - \sqrt{5}}{2}$$

... eigen values

$$n < 2$$

 $\lambda_{\max}$

$$= \frac{-1 + \sqrt{5}}{2}$$


 Space
For
Marks

 Question
No.

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 (Begin answer for each question on a new page)

Q.1

b

Bias-variance decomposition for the expected squared error of a predictor.

If bias is too high, the It means it's underfit. & if variance is too high, it means it's overfit.

If model capacity is too high then no. of layers of the neuron are too much which is consider in variance. it means, it cause high variance which makes the model overfitted. Because of this generalization process of the model get affected.

When the model is overfitted, to overcome this situation, we use methods like early stopping which prevents the model not to get overfitted. Denoising method which will help to reduce noise & reconstruct the model.

To balance bias we use normalization process. If the model is overfitted it means that model contains to many layers of neuron means more no. of noise of each neuron. which fluctuate the flow. and this affects generalization.

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q.2	
	b	
		Autoencoder is a unsupervised model to encode & decode.
		Autoencoder has 3 stages. -
		1) Encode. 2) Bottleneck 3) Decode.
		Overcomplete autoencoder without regularization tends to learn the identity function because it will help to overcome overcompleteness. it will identify the not needed things & will give the balanced things for autoencoder.



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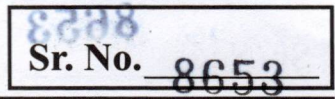
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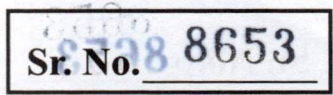


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